

27. Bar Chart of Programming Language Popularity

Aim

To write a Python program to display a bar chart showing the popularity of different programming languages.

Code

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.bar(languages, popularity)

plt.xlabel("Programming Languages")

plt.ylabel("Popularity")

plt.title("Popularity of Programming Languages")

plt.show()
```

Output

A **vertical bar chart** will be displayed with the six programming languages on the X-axis and popularity on the Y-axis.



28. Horizontal Bar Chart

Aim

To write a Python program to display a horizontal bar chart showing the popularity of programming languages.

Code

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

```
plt.barh(languages, popularity)
```

```
plt.xlabel("Popularity")
```

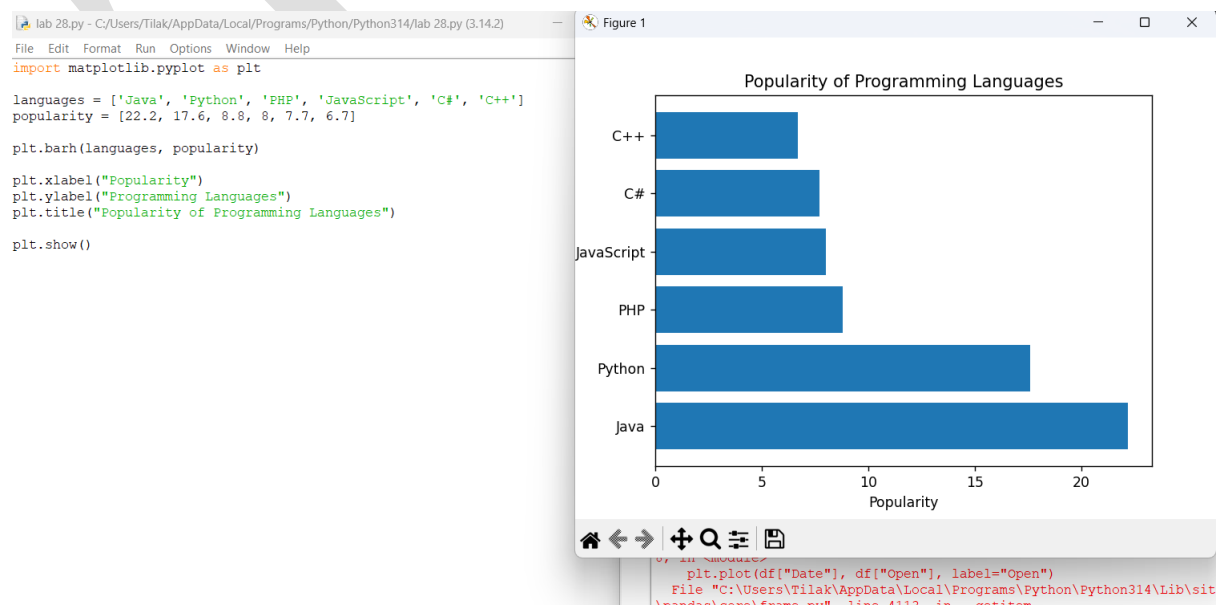
```
plt.ylabel("Programming Languages")
```

```
plt.title("Popularity of Programming Languages")
```

```
plt.show()
```

Output

A **horizontal bar chart** will be displayed, with programming languages on the Y-axis.



29. Bar Chart with Different Colors

Aim

To write a Python program to display a bar chart of programming language popularity using different colors for each bar.

Code

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

```
colors = ['red', 'blue', 'green', 'orange', 'purple', 'yellow']
```

```
plt.bar(languages, popularity, color=colors)
```

```
plt.xlabel("Programming Languages")
```

```
plt.ylabel("Popularity")
```

```
plt.title("Popularity of Programming Languages")
```

```
plt.show()
```

Output

A vertical bar chart will be displayed where **each programming language has a different color**.

```
lab 29.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 29.py (3.14.2)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

colors = ['red', 'blue', 'green', 'orange', 'purple', 'yellow']

plt.bar(languages, popularity, color=colors)

plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")

plt.show()
```

